



FPGA-based 1D-CNN accelerator for real-time arrhythmia classification

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Abstract

Nowadays, cardiovascular diseases are prevalent, real-time arrhythmia detection through electrocardiogram (ECG) is a vital aspect of health monitoring. Consequently, there has been growing interest in wearable edge devices capable of real-time ECG classification. Current convolutional neural networks (CNN) for arrhythmia classification often involve a large number of parameters and have high computational complexity. This work introduces a one-dimensional lightweight convolutional neural network model (LW-CNN), which leverages residual connections and one-dimensional depthwise separable convolution (DSC). The proposed network shows accuracy achieving 99.59% on software-implementation with fewer parameters and lower computational complexity. A model compression method combining unstructured pruning and incremental network quantization (INQ) is implemented to further reduce the model complexity. Additionally, a neural network accelerator based on multiplication-free convolutional processing unit is designed with high level synthesis (HLS) to reduce resource consumption and achieve real-time ECG classification. The entire system is implemented on Xilinx Zynq 7Z020 board leveraging PS-PL synergy design and achieves classification accuracy of 96.55%, a latency of 63 ms under 50-MHZ and a power consumption of 1.78W with resource consumption of 13726 LUT, 9 DSP, and 5.5 BRAM, which improves resource efficiency.

Keywords Wearable edge devices · Convolutional neural network · Arrhythmia classification · Real-time · Resource efficiency

1 Introduction

Arrhythmia is a condition characterized by abnormal heart rhythms or rates. If not diagnosed and treated promptly, it can lead to serious consequences, including heart failure, cardiac arrest, or even sudden death. Recent studies show a significant increase in the incidence of cardiovascular diseases [1–3]. ECG records the heart's electrical activity during each cardiac cycle and displays the data as waveforms [4, 5]. With the growing pressures of modern life and health challenges, there is an urgent need for high-precision devices that can diagnose conditions in real-time. Currently, people often experience delays in obtaining medical diagnoses. In this context, wearable edge devices for continuous ECG monitoring have become an important area of research. Real-time detection ensures timely medical intervention and is crucial for protecting patient health. Therefore, real-time

ECG monitoring is essential for early and accurate arrhythmia diagnosis.

With the rapid advancement of artificial intelligence and neural networks, these technologies are now widely used for classification tasks. Accurate detection and classification of arrhythmias during the initial diagnosis are crucial. Before entering the neural network for classification, the original ECG signals must be preprocessed to minimize external interference. There are various methods for preprocessing ECG signals. For example, Liu Yi proposed a wavelet-based double-threshold ECG signal denoising method in his research [6]. This method improves the accuracy of threshold processing and retains some of the original signal that might otherwise be removed. Meanwhile, Kjell Le developed a technique for detecting and processing high-frequency noise [7], which is used to remove noise from ECG signals. These studies address low-frequency or high-frequency noise in ECG, but they still cannot simultaneously solve the noise problems caused by baseline drift and mains frequency interference.

In recent years, many works have focused on using deep neural networks for feature extraction and classification. CNN have gained prominence due to their strong feature

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extraction capabilities. Numerous ECG classification methods based on CNN have been explored in the literature [5, 8–22]. Among them, [5, 9, 15, 21, 22] primarily use 2D-CNN or a combination of 1D-CNN and 2D-CNN, which always have larger parameter size and higher computational complexity compared to works that use 1D-CNN [8, 10–14, 16–20]. These problems limit their use on resource-constrained edge devices. Additionally, since ECG signals are one-dimensional, 1D-CNN is especially well-suited for ECG classification tasks.

Currently, most existing ECG classification research [5, 8–13] is implemented only in software and relies on GPU. While GPU for neural network computation is generally effective, it often faces challenges such as excessive power consumption. These issues greatly hinder the deployment of classification tasks on wearable edge devices for real-time detection.

Therefore, FPGA's flexibility, low power consumption, and task customization capabilities make it a viable option for implementing neural networks. FPGA can be more easily deployed on wearable edge devices for real-time classification tasks than GPU. To enable real-time monitoring with neural networks, FPGA designs have gained increasing attention [17, 18, 21–28]. Wearable edge devices are characterized by low latency, low power consumption, and limited resources. However, most existing research focuses on improving throughput with hardware architecture design, overlooking the demands of low power consumption and resource limitations.

Given the limited resource on wearable edge devices, model compression techniques show promise for conserving resource. As demonstrated by numerous studies [16, 29–33], model pruning and quantization have become essential strategies for deploying neural networks in resource-constrained environments. Previous works have explored unstructured pruning, quantization and other methods to reduce model complexity. However, these works have not considered the specific application scenarios. Wearable edge devices, constrained by limited resource, demand targeted model compression method to deal with the challenges effectively.

To ensure low latency and low power consumption while addressing the deployment of network for real-time ECG classification on resource-constrained devices, a lightweight 1D-CNN with DSC and residual connections is proposed. This design reduces computational complexity and the number of parameters while maintaining classification accuracy. Building on this network, we propose a model compression method and a multiplication-free convolution processing unit based on FPGA to minimize resource consumption and lower latency. The main contributions of this paper are outlined as follows:

- (a) A lightweight 1D-CNN is proposed in this paper to deal with the substantial number of parameters and high computational complexity. This network leverages residual connectivity and DSC to achieve arrhythmia classification, attaining an accuracy of 99.59% while ensuring lightweight.
- (b) For further lightweight, a model compression method combining INQ and custom unstructured pruning is used. The compressed weights are represented in power of two format with positional encoding.
- (c) A multiplication-free convolutional processing unit is developed for resource efficiency. And a PS-PL synergy system with this processing unit is designed, which not only reduce resource consumption but also achieves low latency and power consumption on wearable edge devices.

2 Related work

2.1 Neural network architecture

In recent years, more and more research has used neural networks to solve the ECG classification problem. Qin [34] proposed a novel neural network architecture for ECG diagnosis based on deformable CNN, which fully exploits the feature-learning capability of deformable convolution to capture both time-domain and lead characteristics of ECG. Hsieh [35] introduced a method based on an end-to-end 1D convolutional neural network architecture. To smoothly deploy neural network models to FPGA-based edge devices, addressing the storage resource limitations of FPGA is an urgent problem. In response, it is crucial to lighten the neural network. Reference [5] achieved excellent classification results on the MIT-BIH dataset by combining 1D convolution and 2D convolution. Hadjer Bechin [9] introduced LC-CNN and fine-tuned MobileNet-V2 models based on 2D CNN, achieving 99.22 and 98.69% accuracy on the MIT-BIH dataset, respectively. These methods for classifying ECG signals using deep learning offer new solutions to the problem. Although they achieve good results, they typically have high computational complexity and cannot be directly deployed to edge devices. Furthermore, [10, 12, 13] used 1D CNN for ECG classification tasks, proving that 1D CNN is effective for classification. Xu [11] proposed a novel 1D CNN structure, achieving an overall accuracy of 99.4% on the MIT-BIH dataset. Additionally, Tesfai [8] introduced a lightweight ShuffleNet structure, enhancing performance by 2% while reducing the parameter count by a factor of nine compared to traditional CNN architectures. These works offer new perspectives for lightweight CNN and inspire our work.

2.2 Model compression

Model compression is essential for reducing computational complexity and parameter size to make the model suitable for FPGA deployment. Qi [32] proposed an interactive neural network compression mechanism that includes out-in-channel pruning and neural network quantization to reduce model complexity. Zhou [29] introduced a novel method to efficiently convert any pre-trained full-precision convolutional neural network model into a low-precision version, with weights constrained to be either power of two or zero. Lu [16] proposed an efficient unstructured sparse CNN accelerator, offering a unique perspective on sparse CNNs for FPGA design. Sui [33] developed an innovative hardware-friendly CNN pruning method called KRP, which prunes the convolutional kernel on a row scale. This method requires much less storage space and helps reduce on-chip hardware resource consumption. These works focus on reducing model and computational complexity. However, using only a single pruning or quantization method does not always yield the desired results. Based on these studies, we propose combining unstructured pruning and quadratic power quantization to compress the one-dimensional convolutional neural network.

2.3 FPGA design

Since the real-time detection and classification of ECG signals are related to the life and health of patients, it is necessary to deploy neural networks on an edge device that can monitor patients in real-time. FPGA has become an essential option for implementing neural networks on edge devices for the past few years due to the difficulty of deploying GPU on edge devices.

Ting [28] developed a method that used a 2D CNN classifier to realize ECG detection and implemented it on the FPGA platform. Guo [36] proposes Angel-Eye, a programmable and flexible CNN accelerator architecture, together with a data quantization strategy and compilation tool. However, these works still have the problem of being difficult to deploy on resource-constrained wearable edge devices. Lu [17] introduced a 1D CNN with a global average pooling layer, achieving 99.13% accuracy, and he also proposed a fully pipelined processing unit array to increase computing efficiency. Such a design makes deploying lightweight networks on wearable edge devices feasible. Rawal [26] focused on hardware optimization techniques based on 1D CNN to reduce computational complexity and power consumption. He proposed Supreme CNN Architecture (SCA) and Software-Selected CNN Architecture (SSCA) to acquire excellent results. Furthermore, Tang [18] proposes a CNN-LSTM architecture developed to enable real-time inference on edge devices, achieving an inference time of approximately 208.2

ms. References [21, 22] proposed hardware architecture schemes for deploying neural networks on FPGA, enhancing the throughput of neural networks on FPGA. These works enable the deployment of neural networks on FPGA; however, they do not focus on further reducing the resource requirements, power consumption, and latency in edge devices by optimizing existing neural network architectures.

Inspired by these studies, we adopted a PS-PL collaborative design scheme based on a multiplication-free convolutional processing unit. Multiplication-free convolution can significantly reduce the demand for DSP and lower latency and power consumption, making them more suitable for wearable edge devices.

3 Methods

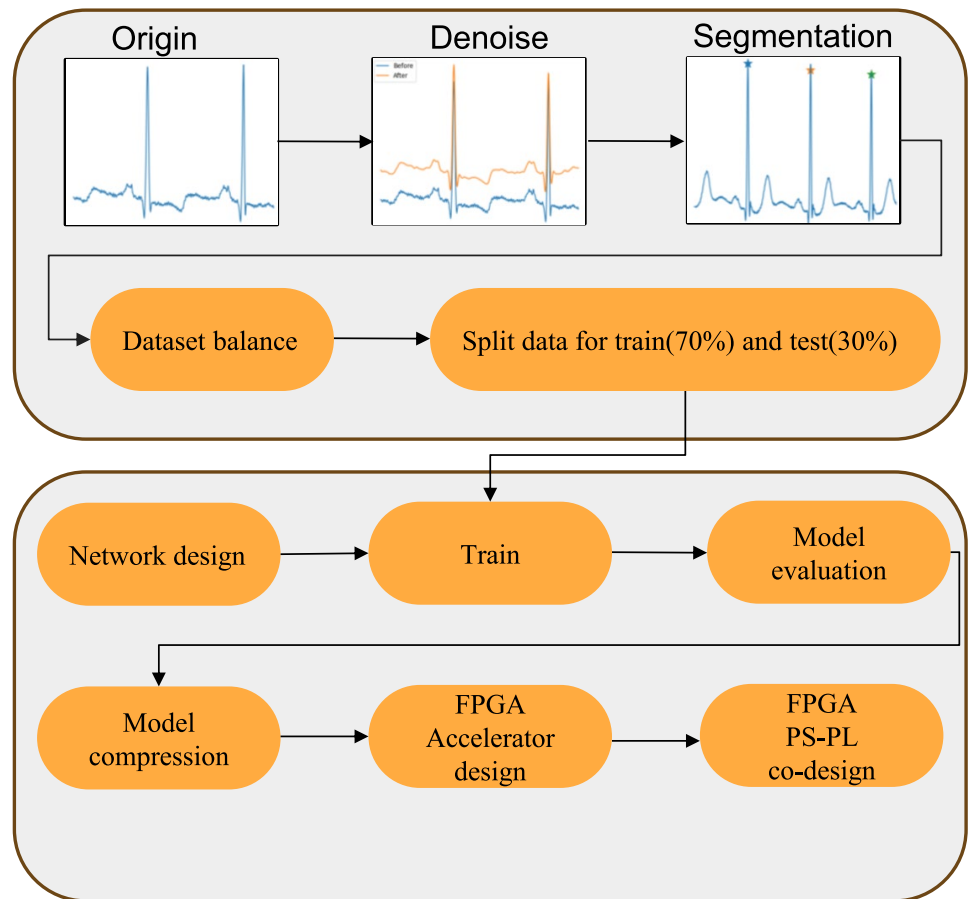
In this section, we initially introduce the dataset and data preprocess method in Sect. 3.1. Following that, we present the overall architecture of our proposed network and model compression method in Sects. 3.2 and 3.3. Finally, we introduce the FPGA design in Sect. 3.4. The workflow is shown in Fig. 1.

3.1 Dataset

The American Association for the Advancement of Medical Instrumentation (AAMI) categorizes arrhythmias into five distinct categories: (a) Category N: normal beats or bundle branch conduction block beats; (b) Category S: supraventricular abnormal beats; (c) Category V: ventricular abnormal beats; (d) Category F: fusion beats; and (e) Category Q: unclassifiable beats. The MIT-BIH dataset utilized in this paper is organized according to the heartbeat categories established by AAMI. Based on the MIT-BIH dataset, we preprocessed it to obtain more distinctive features, and it is as follows:

- (a) We acknowledge the significance of noise reduction in ECG by prior studies [6, 7]. Mitigating excessive noise is essential to enhance the efficiency and accuracy of subsequent classification tasks. This work uses a Butterworth bandpass filter to process the noise in the ECG signals. The filter parameters include a low-pass cutoff frequency of 0.5 Hz, a high-pass cutoff frequency of 30 Hz, an order of 3, and a sampling frequency of 300 Hz, as shown in Table 1. By employing the Butterworth bandpass filter, both baseline drift and industrial frequency interference can be effectively mitigated, ensuring that the processed ECG signals retain their essential characteristics.
- (b) Conventional ECG waveforms typically consist of P waves, QRS complexes, and T waves, with the QRS

Fig. 1 Workflow

**Table 1** Butterworth filter parameters

Low-pass frequency (Hz)	High-pass frequency (Hz)	Order	Sampling frequency (Hz)
0.5	30	3	300

Table 2 Distribution for dataset

Class	Set	Augmented set
N	98,600	98,600
S	2781	98,600
V	7235	98,600
F	802	98,600
Q	33	98,600
Total	109,451	493,000

complexes representing the most critical cardiac activity. In this paper, ECG signals are segmented based on R-wave peak information from the MIT-BIH dataset. Each R-wave peak serves as the central point for segmentation, with 130 sample points taken before the R-wave peak and 129 sample points taken after, resulting in 260 sample points per segment. All samples are stored in a CSV file, with each line containing 260 sample points and the sample type labeled in the last column.

- (c) The total number of samples obtained from the MIT-BIH dataset through the mentioned segmentation method is 109451. However, the dataset looks clearly imbalanced, containing 98600 N-class, 2781 S-class, 7235 V-class, 802 F-class, and 33 Q-class. The SMOTE algorithm is an enhanced approach based on random oversampling. Its primary concept involves analyzing

the minority class samples and generating synthetic samples by interpolating between existing minority class instances. This process helps to address class imbalance, which is particularly important for improving the representation of the minority class in the MIT-BIH dataset. Therefore, We use SMOTE to data augmentation, the provided Table 2 summarizes the significant impact of this augmentation process on the dataset. After the augmentation, augmented dataset was expanded to consist of 98,600 samples in each of the five categories, resulting in a larger dataset comprising 493,000 samples in total, ensuring unbiased evaluation during model training and testing.

3.2 Neural network

This section introduces a one-dimensional network structure for arrhythmia detection to accurately classify ECG while maintaining a lightweight design, as illustrated schematically in Fig. 2.

The previous model based on 2D CNN [9, 21, 22] demonstrated excellent accuracy in ECG processing; however, since ECG are inherently one-dimensional data, converting them to two-dimensional space introduces greater computational complexity and results in larger parameters. These works make deployment on edge devices for real-time monitoring less feasible. In order to facilitate deployment on edge devices, the network architecture proposed in this paper is designed to be lightweight and hardware-friendly. The network comprises seven residual blocks, all convolution layers employing one-dimensional convolution. DSC is utilized in place of traditional convolution layers for feature extraction to reduce parameters, lower computational complexity, and improve classification performance. To further minimize model complexity, the bias in the convolutional layers is omitted. Subsequently, the final classification is performed through three fully connected layers, which categorizes the output into five classes.

DSC combines channel-by-channel convolution with pointwise convolution, as illustrated in Figs. 3 and 4. Initially, a one-dimensional convolution kernel processes each input channel independently as shown in (1), where i is the index of points in the feature map, $Y_{i,c}$ is the result of one channel, $X_{i+m,c}$ is the input data at position $i + m$ and c is the

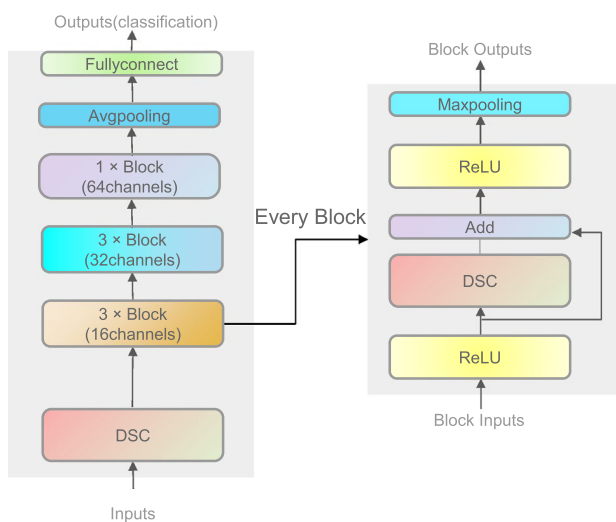


Fig. 2 LW-CNN

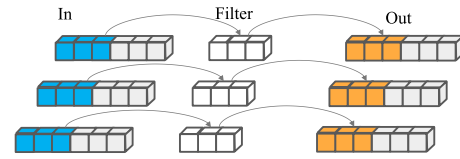


Fig. 3 Depthwise convolution

index of the feature map channel, $W_{m,c}$ is the weight at position (m, c) , K is kernel size.

$$Y_{i,c} = \sum_{m=0}^{K-1} X_{i+m,c} \cdot W_{m,c} \quad (1)$$

Then, a 1×1 kernel performs pointwise convolution to fuse all channels, where $Z_{i,c_{out}}$ represents the result of one output channel at position i , $X_{i,c}$ is the input data at position i and c is feature map channel, and $W_{c,c_{out}}$ is the weight at channel (c, c_{out}) . Therefore, the various channels of the feature map can be better fused, which is calculated as:

$$Z_{i,c_{out}} = \sum_{c=0}^{c_{in}-1} X_{i,c} \cdot W_{c,c_{out}} \quad (2)$$

LW-CNN employs residual blocks to enhance the neural network's performance and prevent gradient vanishing during training. In the design of each residual block, the input features are first processed by a nonlinear ReLU activation layer to extract the nonlinear characteristics of the data. The output from this ReLU layer then feeds into a DSC layer. Residual connections are introduced between each residual block's input and the convolutional layer's output, with an additional convolutional layer for transforming the channel if necessary. Residual connections are shown as (3), where $F(X)$ is the output after multiple layers computation and y is the result.

$$y = F(x) + x \quad (3)$$

The design of the residual connectivity for the blocks is represented in (4). where $Block_{Input}$ is the input data of the residual block. $Block_{Input}$ enter the first ReLU layer firstly, and then the result which processed by DSC is added with

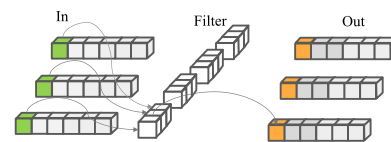


Fig. 4 Pointwise convolution

the original $Block_{Input}$. The output through the second ReLU layer to further enhance the model's representational capacity and is ultimately directed into the Maxpooling layer to reduce the dimensionality of the input features while extracting local features.

$$Block_{out} = \text{Maxpooling}(\text{ReLU}(\text{Block}_{Input} + \text{DSC}(\text{ReLU}(\text{Block}_{input})))) \quad (4)$$

The first three residual blocks' input channels and output channels are both 16; the fourth residual block's input channels are 16 and output channels are 32; the fifth and sixth residual blocks maintain input channels and output channels are both 32; finally, the seventh residual block's input channels are 32 and output channels are 64.

To demonstrate the effectiveness of depth-wise separable convolution in reducing parameters, we compare the number of parameters of standard convolution and depth-wise separable convolution. Equations (5) and (6) respectively show the number of parameters of standard convolution and depthwise separable convolution, K represents the kernel size, C_{In} and C_{Out} represent the number of input channel and output channel. From (5) and (6), it can be obtained that the total number of parameters in the LW-CNN architecture is 11,736, and the number of parameters in each part is shown in Table 3. In contrast, if traditional convolution were employed, the total parameter count would rise to 22,037 for the same network structure.

$$P_{Standard} = K \times C_{in} \times C_{Out} \quad (5)$$

$$P_{DSC} = K \times C_{in} + C_{in} \times C_{Out} \quad (6)$$

This comparison clearly demonstrates that using 1D-DSC significantly reduces the number of parameters only 53.3%

Table 3 Parameters

Part	Kernel size	Parameters
1D-DSC	3	19
Block1	3	912
Block2	3	3312
Block3	3	4192
Linear	–	3301
All	–	11736

of those required for traditional convolution resulting in a reduction of more than 46%. This reduction is crucial for achieving a lightweight network, which is essential for real-time edge device monitoring.

3.3 Model compression

This subsection introduces the model compression method. This approach aims to create a less complex network model which is more suitable for deployment on wearable edge devices, facilitating real-time monitoring and further reducing hardware demands.

The pruning strategy in this section adopts unstructured pruning of Pytorch and calculates the MASK matrix through a custom method. As illustrated in (7), x represents each element in the convolution kernel. We only retain the element with the largest absolute value in each kernel as the non-zero weight w and set the elements at other positions 0. The weights pruned by such a method can reduce the parameters by more than 60%.

$$W_i = \max_{j=1,2,3} |x_j| \quad (7)$$

In addition, a position code is set according to the position of the retained weights, which is used to skip the redundant zero-weight multiplication calculation, and the whole pruning process is shown in Fig. 5. After pruning, the information of each kernel is changed from the original three parameters to one parameter and one position encoding. The number of parameters which can be learned is reduced to one-third of the original one, reducing the model complexity.

The quantization strategy uses INQ algorithm, which quantizes the model weights into quadratic power and is represented by 8 bits. The primary approach uses group-wise quantization firstly and then freezing the weights of each quantized group to ensure these parameters are no longer involved in subsequent training. Then, we will quantify the weights group by group according to the divided groups, and after quantization, the accuracy will be recovered by retraining until all the subgroups are quantized. After quantization, the quadratic power weights in each kernel are saved in the form of the number of shift operation needed; the result is shown in Table 4.

3.4 FPGA design

This section mainly introduces the design of accelerator IP cores in HLS. The primary considerations are resource consumption, latency and power consumption to realize real-time

Fig. 5 Pruning process

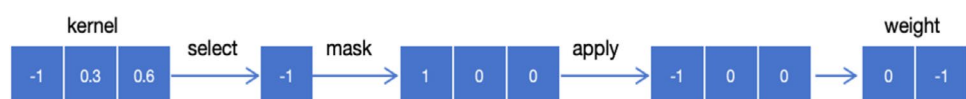


Table 4 Quantified weights

Quantified-weight	Saved in files
2^{-1}	1
2^{-2}	2
2^{-3}	3
..	..

ECG classification on wearable edge devices. We propose a multiplication-free convolution processing unit to address the problem of resource constraints and real-time requirements of edge devices. In the convolutional layer IP core, we use the method of skipping zero-value calculations and shifting to reduce resource consumption, latency and power consumption. In addition, considering that parameters and operating mode of each layer in the neural network are different, a configurable interface is added to the IP core. The DSC block in the network proposed include two types of convolution: depth-by-depth convolution and point-by-point convolution. Therefore, the convolutional layer IP core is designed to be configured to change parameters, such as kernel size, mode of performing convolution, padding, number of input channels, number of output channels, and input length. In addition, the ReLU layer, the fully connected layer, and the pooling layer all use this design approach to achieve the generality of IP cores.

Figure 6 illustrates the depth-by-depth convolution operation in mode 0. Since each channel performs independent computation and does not interfere with each other, we design a sliding window for the convolution in each input channel. According to the position encoding in the weight file, one element is selected within this sliding window; and since the weights have been quantized in powers of two, shifting can be used instead of the traditional multiplication

in the computation. Furthermore, we optimize the outermost channel loop through HLS optimization instructions to achieve efficient parallel computing for multi-channel, such as the 16-channel parallel and 32-channel parallel we have used.

Figure 6 also illustrates the process when performing point-by-point convolution operation in mode 1, where we need to use a kernel of size 1 to establish the link between different channels. Therefore, we select valid output elements according to the two layers of loops of the outermost firstly, then perform the shift calculation by selecting elements at the current position on each input channel. Finally, the result is obtained by accumulation. Once we finish processing the outermost output channel in the loop, we get the results for all the output channels.

Since designing exclusively with PL can lead to issues such as reduced flexibility and imbalanced resource utilization. PS-PL synergy is used to realize the deployment of neural networks, enhance flexibility, and make the whole system more reliable.

Figure 7 shows the PS-PL synergy architecture, in which the accelerator IP cores are mounted with the configured ARM processor. The system uses the efficient AXI bus for communication in this design. Firstly, the ARM is mainly responsible for the data communication and process control of the entire system by sending directives. Then, the SD card is used to store the sample data and weight files for the LW-CNN, and the DDR is used to cache the data read from the SD card. In the software design of the ARM, the ARM sends directives to the PL-side accelerators to execute each step of the computation according to the model structure of the LW-CNN. At the same time, it ensures that the corresponding parameters

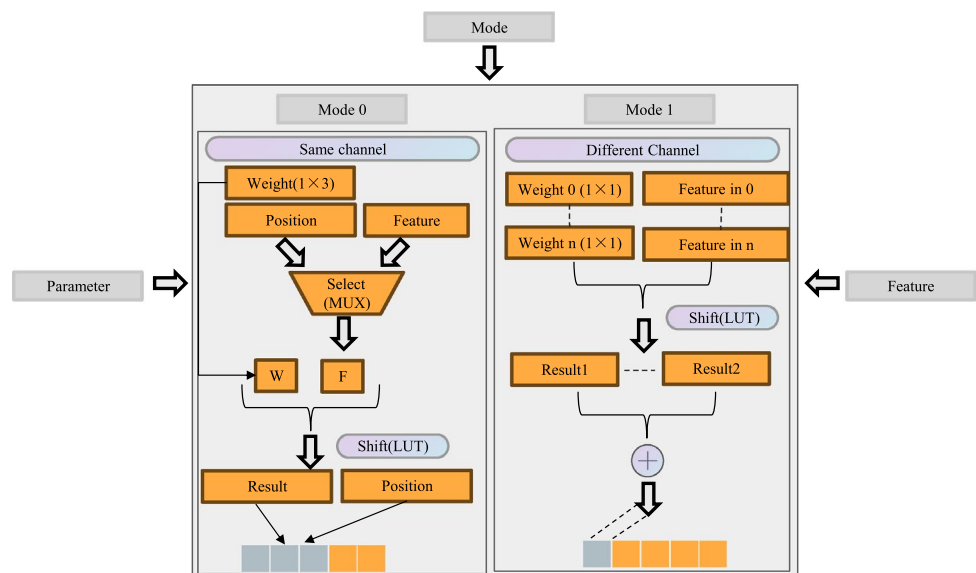
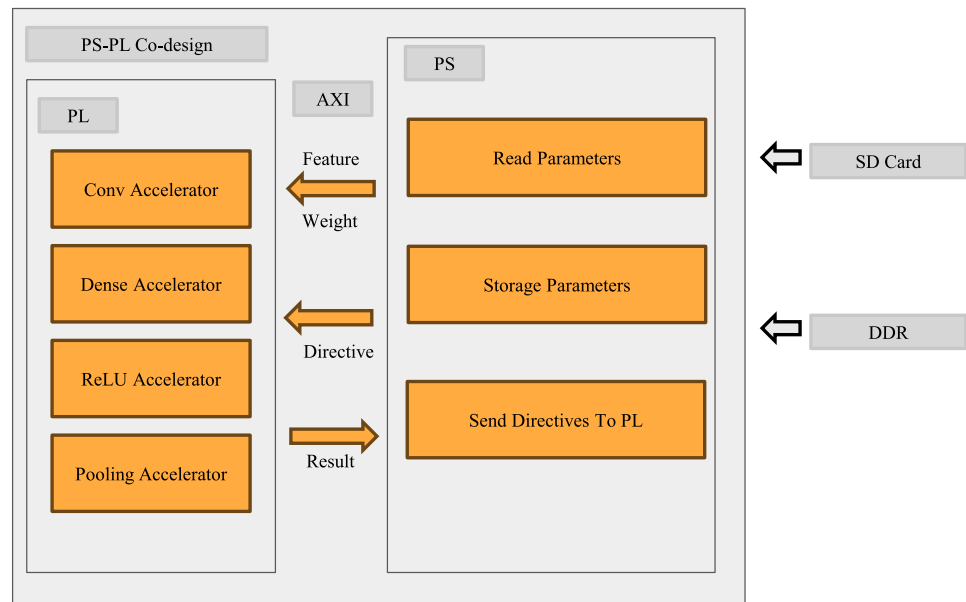
Fig. 6 Configurable convolution layer accelerator design

Fig. 7 PS-PL co-design

and data can be correctly transferred to the accelerator via DMA before the calculations begin. Through this PS-PL synergies approach, we can accurately control the computational process of the neural network and ensure the correctness and efficiency of the obtained inference results.

4 Experiments

4.1 Setup

To validate the performance of the 1D CNN proposed, we use the preprocessed MIT-BIH dataset to test the network's performance on the arrhythmia classification.

The neural network proposed is built and trained under the PyTorch. The dataset was split into two parts: 70% for training, and 30% for testing. Since this neural network is to accomplish a multi-classification task, the cross-entropy loss is chosen for the loss function. In addition, we added dropout with a rate of 0.5 in the fully connected layer to prevent possible overfitting during the training process; finally, the learning rate was set to $1e-3$, and the parameters were updated using the adaptive learning rate optimizer (Adam) during the training and training epochs set to 30.

After compression, it is imperative to fine-tune the model through re-training to restore its accuracy. During the retraining phase, learning rate set to $1e-2$. Loss function continues to be the cross-entropy loss, which is appropriate for multi-class classification tasks. The optimization algorithm transitions from the Adam to a customized stochastic gradient descent (SGD) optimizer. We use Adam

for pretraining due to its superior convergence properties, while retraining benefits more from the stability provided by a fixed learning rate. Consequently, SGD is chosen for retraining to achieve optimal performance. Moreover, the customized SGD incorporates the grouping information necessary for quantization, enabling the freezing of quantized parameters during the re-training phase.

4.2 Network result

The experiments demonstrate that the model achieves high accuracy while preventing overfitting because of the design of residual connections and dropout, which enable the model to perform exceptionally well on both the training and test sets. Ultimately, it achieves 99.59% accuracy on the test set, confirming that LW-CNN can effectively and reliably execute the ECG classification task.

To further evaluate the performance of LW-CNN, we analyze its ability to classify each specific category of arrhythmia using the confusion matrix, as shown in Fig. 8. The confusion matrix reveals balanced and efficient classification performance across all categories. LW-CNN achieves the highest classification performance of 100% in category 4 (class Q), followed by more than 99% accuracy in categories 1 (class S), 2 (class V), 3 (class F), and finally, over 98% accuracy on category 0 (class N).

In this section, we also provide a comprehensive comparison between the network structure LW-CNN proposed in this paper and those in previous works regarding the number of parameters and classification performance. In addition, we compared the performance of the different network structures by Precision, Recall, and F1 score, as shown in (8)–(10),

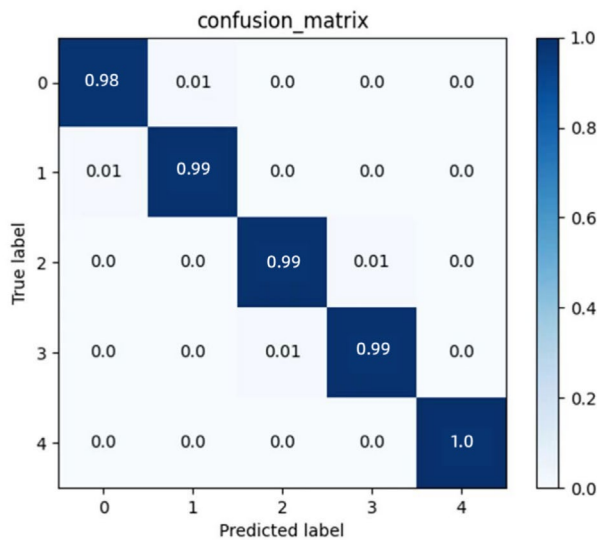


Fig. 8 Confusion matrix

respectively, which are commonly used evaluation metrics in classification tasks.

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (8)$$

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (9)$$

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{(\text{Precision} + \text{Recall})} \quad (10)$$

Based on the preprocessed MIT-BIH dataset, the comparison results of the proposed network with the lightweight ShuffleNet [8], the CNN based on coupled convolution structure and dropout mechanism [11], and other software-implemented networks, in terms of parameter scale and overall accuracy, are shown in Table 5. The results comparing the important evaluation metrics of the networks from [8] and [11] are displayed in Table 6.

Comparisons above show that LW-CNN reduces the number of parameters by approximately 1000 compared to

Table 5 Comparison on network accuracy and parameters

Work	Method	Parameters	Accuracy (%)
Tesfai [8]	Shufflenet	9718	92.23
Tiwari [14]	1D-CNN	29,263	95.6
Dutta [12]	1D-CNN	–	96.83
Lu [17]	1D-CNN	11,065	99.13
Xu [11]	1D-CNN	–	99.4
Proposed model	1D-CNN	11,736	99.59

Table 6 Comparison of performance indicators

Work	Method	Class	Precision	Recall	F1-score
H. Tesfai [8]	1D-ShuffleNet	N	1.00	0.99	1.00
		S	0.96	0.91	0.93
		V	0.99	0.99	0.99
		F	0.97	0.96	0.97
		Q	–	–	–
X. Xu and H.Liu [11]	1D-CNN	N	–	–	–
		S	0.991	0.975	–
		V	0.999	0.992	–
		F	–	–	–
		Q	–	–	–
Proposed model	1D-CNN	N	0.99	0.98	0.987
		S	0.98	0.994	0.989
		V	0.99	0.99	0.99
		F	0.994	0.989	0.991
		Q	1	1	1

previous lightweight ShuffleNet [8], while accuracy achieved 1.5% improvement. The experiments also indicate that compared to the CNN in [11], using DSC and residual connections reduces most parameters and improves accuracy by 0.2%. Additionally, the proposed model shows better classification performance for certain specific classes. Compared to [17], accuracy improves by 0.4%. Furthermore, due to the application of deeper networks and more separable convolutional layers, the classification performance for the S and F classes shows a significant improvement of 2–8% compared to [8]. These results indicate that CNN incorporating residual connections and DSC can efficiently and effectively execute ECG classification task.

4.3 FPGA result

In this section, we chose the Zynq-7Z020 platform due to its mature development environment, balancing energy consumption and performance. First, we conducted several classification simulation experiments on the Zynq-7Z020 development board using the HLS platform with different test set samples to verify its correctness. Then, we set up an SoC platform to analyze resource utilization, validating the feasibility of deploying the LW-CNN neural network on edge devices and highlighting its hardware-friendly feature of low resource consumption. Finally, the compressed model was deployed on FPGA to demonstrate its low-latency and low-power characteristics. To verify the correctness of the designed IP cores, we conducted multiple simulation experiments in HLS, and in each case, we obtained correct classification results that were consistent with the results computed in Pytorch.

Then, we exported the accelerator modules designed in HLS and completed the construction of the Zynq-based SoC platform in Vivado 2023. Finally, we obtained the resource utilization of the Zynq-7Z020 through Vivado.

In terms of resource utilization, it can be observed that, due to the use of shifting instead of multiplication in the convolutional layer, the convolutional layer computation utilizes the more abundant LUT resources rather than the scarce DSP resources. Since other layers' computational complexity is more limited than the convolutional layer, DSP resources are still used for operations outside the convolutional layer. DSP consumption accounted for only 4.09% of the total on-board DSP resources. Thanks to the lightweight network design, other resources, such as BRAM, are also used in small proportions, significantly reducing the hardware resource requirements compared to previous works [17, 18, 26]. The experimental results show that the convolutional layer without multiplication, based on the HLS design in this paper, has effectively reduced on-chip DSP resource consumption. Table 7 compares the LW-CNN with other methods in terms of resource consumption. The results indicate that LW-CNN occupies fewer resources across all aspects and is easier to deploy on resource-constrained FPGAs for real-time ECG signal classification. The overall accuracy of the compressed model on the FPGA is 96.55%, demonstrating its ability to efficiently and accurately complete the ECG classification task. Additionally, the latency of LW-CNN on Zynq-7Z020 and RTX4060 was compared, as well as the latency in [18]. On the other hand, in Table 7, we compare the deployment of LW-CNN with [22] on FPGA, and in Table 8, we also compare the performance of [22] and LW-CNN on different platforms across various metrics. It is evident that 2D VGG-16 consumes more hardware resources, requires longer execution time, and has higher power consumption. Therefore, 1D CNN outperforms 2D CNN in these aspects.

We deploy the compressed LW-CNN on FPGA, leveraging its lightweight nature and further compression, along with a multiplication-free convolutional processing unit that simplifies multiplications into shift operations. Therefore, LW-CNN can be efficiently deployed on resource-constrained wearable edge devices, achieving a latency of 63 milliseconds and power consumption of 1.7 W at a clock frequency of 50 MHz while reducing resource requirements. Compared to previous work, our approach reduces resource demands and achieves lower latency and power consumption, which is significant for real-time classification of ECG signals on edge devices.

Table 7 Comparison of performance indicators

Work	Device	Method	Accuracy (%)	LUT	FF	DSP	BRAM	Architecture	CNN size (GOP)	Power (W)	Latency (ms)
Rawal [26]	Zynq-ZCU106	1D-CNN	86.37	184,450	57,855	0	251	PS-PL	-	0.628	-
Tang [18]	Xilinx Zynq-7000	CNN-LSTM	98.63	36,708	36,176	130	63	PS-PL	-	-	208.2
Gong [19]	Zynq-XC7Z020	LeNet-5	-	38,136	42,618	205	242	PL	5.3×10^{-4}	2.15	-
Jaramillo [20]	Artix-7	1D-CNN	94	128,960	2080	121	16.5	PL	-	-	-
Aimar [21]	Zynq-7100	VGG-16	-	229,000	-	128	386	PL	-	-	-
Chang [22]	ZCU102	VGG-16	-	78,236	52,088	655	1573	-	-	24.3	101.326
Lu [17]	Zynq-ZC706	1D-CNN	99.13	1538	-	80	12	PL	1.028×10^{-3}	-	-
Proposed	Zynq-XC7Z020	1D-CNN	96.55	13,726	18,379	9	5.5	PS-PL	4.66701×10^{-4}	1.78	63

Table 8 Comparison of different platform

Work	Device	Methods	Power (W)	Latency (ms)
Chang [22]	i7 7700	VGG-16	67	930.08
	RTX2080Ti	VGG-16	293	14.74
	ZCU102	VGG-16	24.3	101.326
Proposed	RTX4060	1D-CNN	–	179.2
	Zynq-XC7Z020	1D-CNN	1.78	63

5 Conclusions and future work

This paper introduces a lightweight 1D-CNN based on residual concatenation and DSC. Experiments on the MIT-BIH dataset show that proposed network effectively classifies ECG arrhythmias. Compared with the network models in previous works, the performance of this network model has improved by approximately 0.1–4%, and it achieves an accuracy rate of 99.59% with fewer trainable parameters. After applying the proposed model compression strategy, the network demonstrates reliable classification ability. Achieving real-time classify with lower resource utilization when deployed on an FPGA. In the future, the proposed neural network classifier can also be applied to other scenarios that require to classify and have strict limitations on the parameters. Additionally, the purpose of using HLS in this paper is to achieve rapid verification, however, using Verilog to design the computing module on the PL may achieve better throughput. Based on current research, we will focus on using Verilog for further optimization, such as optimize the hardware architecture.

Author Contributions L. wrote the text of the main manuscript and Z.L.W. offers guidance on article writing. All authors reviewed the manuscript.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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